



Evaluation 1: Duolingo XR — Immersive Chinese Learning

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Objective and Validation Metrics

The objective of Interactive Prototype 1 was to test whether beginner Chinese learners could understand and use an object-based vocabulary learning interaction inside a virtual Study Room. The core assumption being tested was that connecting Chinese vocabulary to recognisable 3D objects, text labels, and pronunciation audio would help beginner learners understand and remember the relationship between a Chinese word and its meaning.

The test aimed to determine:

- Whether participants could discover that objects were clickable to trigger vocabulary learning.
- Whether the vocabulary card's Chinese character, pinyin, and English translation were clear and useful.
- Whether pronunciation audio helped participants remember new vocabulary.
- Whether participants could correctly associate words with the right 3D objects.
- Whether navigation and object selection were intuitive enough not to interfere with learning.

The original Duolingo XR concept follows a three-stage learning flow: (1) Learn — object-vocabulary association, tested here; (2) Recognition — removing English support and testing whether learners can identify objects from Chinese/pinyin alone; and (3) Action — understanding and executing spoken commands such as "请拿起书" (please pick up the book), using grab, move/place, and spatial XR interactions. Prototype 1 deliberately narrows its scope to Stage 1 only, since this stage needed validation before recognition and action could be meaningfully layered on top.

Success was defined as:

- Most participants discovering object selection with minimal help.
- Most participants opening and explore all five vocabulary objects.
- Participants would correctly recall at least 2 out of 3 tested vocabulary-object relationships.
- Positive ratings (4-5/5) for the vocabulary card and audio.
- No navigation or UI issues blocking the core learning task.

These metrics combined quantitative data (hints, objects opened, recall accuracy, ratings) with think-aloud observations, assessing not just task completion but genuine understanding.

Results

Four participants completed the test. The original target was five, but time did not allow for a fifth session. Three participants had no previous Chinese experience. One participant had some previous experience.

Participant	Chinese Experience	First Interaction	Hints	Objects Opened	Correct Recall	Interaction /5	UI /5	Audio /5	Learning Time
P1	None	10 sec	2	5/5	2/3	5	5	5	2:03
P2	None	17 sec	3	4/5	1/3	5	5	5	1:37
P3	Some	9 sec	1	5/5	3/3	5	5	5	1:00
P4	None	15 sec	4	5/5	2/3	4	4	5	1:21

All four required at least one hint before their first successful interaction. Time to first interaction ranged from 9–17 seconds; hints ranged from 1–4 per participant.

Three participants opened all five objects (5/5); one opened four (4/5). All four discovered the Listen button and used it frequently (11–18 times each).

Recall accuracy ranged from 1–3 out of 3: one participant scored 3/3, two scored 2/3, one scored 1/3.

Interaction ease and vocabulary UI usefulness were rated 5/5 by three participants and 4/5 by one. Audio usefulness was rated 5/5 by all four. Learning time ranged from 1:00 to 2:03.

Several participants said they were unsure what to do at first, with no initial instructions. One did not realise the right mouse button controlled looking around. One said the vocabulary card's colour blended into the background.

Suggested additions included sound effects, prompts, wall/floor instructions, and a slower pace. Most participants said connecting words to objects helped them remember, though one noted more practice was still needed.

Analysis

Insight 1 — Lack of onboarding made the controls and interaction difficult to discover

The clearest usability issue was the lack of instructions when participants first entered the Study Room. All four participants required at least one hint, and several participants said that they were unsure what to do at the beginning. Some users did not know that they needed to hold the right mouse button to rotate the camera, while others did not immediately understand that clicking a 3D object would open its vocabulary card.

This suggests that the interaction itself was functional, but its **discoverability was weak**. Users should not need verbal instructions from the facilitator before they can begin exploring. In the next iteration, an onboarding guide should be integrated into the environment, for example:

- **WASD = Move**
- **Hold Right Mouse Button + Move Mouse = Look Around**
- **Left Click = Select an Object**
- **Q / E = Down / Up** as a planned vertical-movement control

The movement keys could be displayed on the floor, while camera and object-selection instructions could appear on a wall or as a temporary screen prompt. This would allow users to understand the interaction independently and reduce the number of facilitator hints.

Insight 2 — Audio was highly effective, but stronger audio feedback could make learning more engaging

Pronunciation audio was one of the strongest features of the prototype. All four participants rated the usefulness of audio **5/5**, and participants repeatedly used the pronunciation function during the learning task. Several participants also described audio as helpful for connecting the Chinese word with the corresponding object.

This indicates that audio should remain a central part of the learning experience. However, the current prototype only uses audio to pronounce vocabulary. In a future recognition task, audio could also provide immediate feedback after the learner selects an object. For example, a correct choice could trigger positive feedback such as **“Great!”** (真棒!), while an incorrect choice could use supportive feedback such as **“Try again”** (再来一次) or **“Keep trying”** (继续努力).

Adding correct and incorrect sound feedback could make the experience feel more interactive and game-like. It would also provide users with immediate confirmation without requiring the facilitator to explain whether an answer was correct. This would better support the broader Duolingo XR concept of learning through repeated interaction and feedback.

Insight 3 — The vocabulary card was useful, but it needs stronger visual emphasis

Participants generally rated the vocabulary interface positively, with ratings between **4/5** and **5/5**. However, one participant specifically commented that the vocabulary card colour was too similar to the overall Study Room environment, making it difficult to distinguish from the background.

This suggests that the information shown on the card was useful, but the card's **visual hierarchy and contrast were not strong enough**. Because the vocabulary card contains the main learning information, it should immediately attract the learner's attention after an object is selected.

The next iteration should use a more contrasting card background, increase the size and visual emphasis of the Chinese character, and create clearer separation between the Chinese word, pinyin, English translation, and pronunciation control. The audio button could also use a clearer speaker icon. These changes would make the card easier to notice and read while keeping the user's attention on the relationship between the vocabulary and the selected 3D object.



Evaluation of Aims

The main aim was to test whether users could learn Chinese vocabulary by connecting words with 3D objects.

This aim was partly validated. Most participants could open the learning objects and understand the vocabulary cards. Three out of four participants correctly recalled at least 2 out of 3 tested words. The audio was also rated very positively.

However, the interaction was not fully clear at the beginning. All participants needed at least one hint. Some users did not know how to look around or that they needed to click objects.

The original ideas about reduced English support, object recognition, and physical actions were not tested in this prototype. These aims remain for future development.

Concept Iteration

Based on the testing, the next version will include three main changes.

First, I will add clearer onboarding instructions. Users will be shown how to move, look around, and select objects. This should reduce confusion when they first enter the Study Room.

Second, I will add more physical interaction. Users will be able to grab and move virtual objects. This will make the experience closer to the original XR concept and connect language learning with physical actions.

Third, I will improve audio feedback. Correct and incorrect choices can trigger short responses such as “Great!”, “Try again!”, or “Keep going!”. This can make the learning experience more interactive.

Finally, I will redesign the vocabulary card. The card will use stronger contrast, larger Chinese text, and a clearer audio button. This should make the learning information easier to notice and read.

Reflection and Next Steps

This testing session showed that the prototype was useful for testing the basic idea, but it was still very rough and incomplete.

I originally planned to test five participants, but I only completed four tests because of time limits. In the next testing session, I need to manage the testing time better and prepare the process more efficiently.

The prototype also had very limited guidance. When participants first entered the Study Room, several of them were unsure what to do. They did not immediately know how to move, change the camera view, or click objects to open vocabulary cards. This showed that the prototype needed clearer onboarding and instructions.

The current version also only tested a small part of the original concept. It mainly focused on vocabulary learning and object selection. In the next prototype, I want to improve the visual quality, add clearer instructions, include more physical interaction such as grabbing objects, and add stronger feedback for correct and incorrect actions.

For future testing, I will also try to test more participants and give each participant a clearer but consistent testing process.

GitHub Link

https://github.com/yenyong1211/YININGZHU_DECO7230_2026/tree/main/Prototype1

Appendix

Appendix A — Participant 1 Raw Data

Measure	Data
Chinese experience	None
Time until first successful object interaction	10s
Needed interaction hint	Yes 2
Objects successfully opened	5/5



Listen button discovered	Yes
Pronunciation used	18
笔 recognised correctly	Yes
书桌 recognised correctly	Yes
电脑 recognised correctly	No
Navigation difficulty	Notes
Interaction ease	5
Vocabulary UI usefulness	5
Audio usefulness	5
Participant comments	Add some prompts and make it more interactive.
Learn time	2min3s

1. **Did connecting the word to a 3D object help you remember it? Why or why not?** Yes.
Having a physical object can help with memorization, but I still think you need to practice more.
2. **Was anything confusing about the vocabulary card or controls?** At first, I was a little lost—I didn't know what to do, and there were no instructions.
3. **What would you change or improve?** You can add some interactive elements, sound effects, and prompts.

Appendix B — Participant 2 Raw Data

Measure	Data
Chinese experience	None
Time until first successful object interaction	17s
Needed interaction hint	Yes 3
Objects successfully opened	4/5
Listen button discovered	Yes
Pronunciation used	13
笔 recognised correctly	No
书桌 recognised correctly	Yes
椅子 recognised correctly	No
Navigation difficulty	Notes
Interaction ease	5
Vocabulary UI usefulness	5
Audio usefulness	5
Participant comments	Not sure what direction to take
Learn time	1mins37s

1. **Did connecting the word to a 3D object help you remember it? Why or why not?** Useful.
Real-life objects and audio make learning Chinese easier.
2. **Was anything confusing about the vocabulary card or controls?** Not sure what direction to take.



3. **What would you change or improve?** The color of the card is too similar to the overall color scheme, making it hard to distinguish. You could try a different color.

Appendix C — Participant 2 Raw Data

Measure	Data
Chinese experience	Some
Time until first successful object interaction	9s
Needed interaction hint	Yes 1
Objects successfully opened	5/5
Listen button discovered	Yes
Pronunciation used	11
笔 recognised correctly	Yes
书桌 recognised correctly	Yes
电脑 recognised correctly	Yes
Navigation difficulty	Notes
Interaction ease	5
Vocabulary UI usefulness	5
Audio usefulness	5
Participant comments	Notes
Learn time	1mins

1. **Did connecting the word to a 3D object help you remember it? Why or why not?** Yes, especially for simple objects like the book and chair. I could remember the word by remembering where the object was in the room.
2. **Was anything confusing about the vocabulary card or controls?** I don't know what to do when I open the interface.
3. **What would you change or improve?** You could provide some hints, such as placing instructions on the wall about how to turn, placing movement keys on the floor, and explaining how to click on objects.

Appendix D — Participant 2 Raw Data

Measure	Data
Chinese experience	None
Time until first successful object interaction	15s
Needed interaction hint	Yes 4
Objects successfully opened	5/5
Listen button discovered	Yes
Pronunciation used	17
笔 recognised correctly	Yes
书桌 recognised correctly	No
电脑 recognised correctly	Yes
Navigation difficulty	Notes



Object-selection errors	Number
Interaction ease	4
Vocabulary UI usefulness	4
Audio usefulness	5
Participant comments	Add some prompts and make it more interactive.
Learn time	1mins21s

1. **Did connecting the word to a 3D object help you remember it? Why or why not?** Yes. Seeing the Chinese word while looking at the actual object made it easier to create a connection in my memory.
2. **Was anything confusing about the vocabulary card or controls?** The vocabulary card was clear, but I was not sure at first that I needed to hold the right mouse button to look around.
3. **What would you change or improve?** She suggested keeping the pace slow to give them time to think and learn.